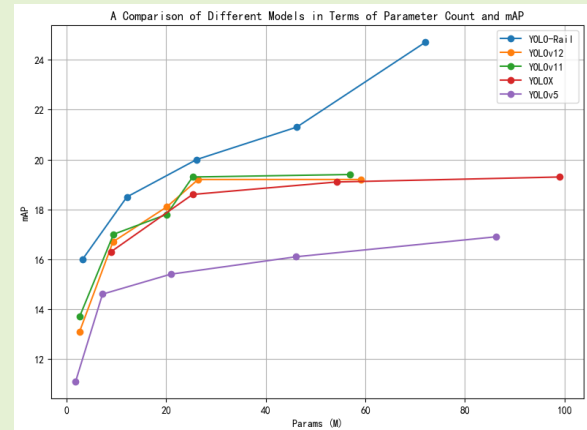


YOLO-Rail: An Improved YOLO Model for Obstacle Detection on Railway Tracks

Zhanyang Wang, Xin Du

Abstract—Automatic detection of obstacles on railway tracks is of great significance for ensuring train operation safety. This paper proposes YOLO-Rail, a lightweight YOLO-based object detection model designed for real-time railway track obstacle detection. The model integrates ConvNeXtV2 modules and depthwise separable convolutions to improve feature representation, and employs a feature fusion network combining PAN and FPN to enhance multi-scale feature integration and detection accuracy. First, the model is pre-trained on the large-scale generic object detection dataset COCO to acquire stronger feature generalization ability. Subsequently, the pre-trained weights are transferred to a railway track obstacle dataset for fine-tuning, effectively addressing the issue of limited scene-specific data. Experimental results demonstrate that YOLO-Rail achieves superior detection accuracy and inference speed compared to existing mainstream object detection algorithms in railway obstacle detection tasks. It significantly enhances detection robustness and reliability while ensuring real-time performance. This study provides a feasible solution for applications in railway safety monitoring and intelligent transportation systems.

Index Terms—Railway Track Obstacles, Real-time Detection, Safety Monitoring, Transfer Learning, YOLO-Rail



I. INTRODUCTION

WITH the continuous expansion of railway transportation networks and the increasing speed of train operations, railway operational safety has become a growing concern. Foreign objects, fallen debris, or other obstacles on the tracks can easily trigger transportation accidents, leading to significant economic losses and casualties [1]–[3]. Therefore, how to efficiently and accurately detect obstacles on railway tracks has become an important research topic in the fields of intelligent railways and intelligent transportation systems.

In recent years, deep learning has achieved breakthrough progress in the field of computer vision, particularly with single-stage object detection algorithms represented by the YOLO series [4]. Thanks to their end-to-end detection pipeline, superior real-time performance, and high detection accuracy, these methods have been widely applied in autonomous driving, video surveillance, industrial inspection, and other scenarios [5]–[8]. However, existing object detection algorithms still face several challenges in railway track environments: First, the complex background and varying lighting

conditions in railway scenes often lead to false positives and missed detections in conventional detection models. Second, the wide variety of obstacles on railway tracks, many of which are small in size, pose difficulties for current detection algorithms in terms of small object detection. Third, the limited scale of existing railway obstacle datasets often results in poor generalization ability when models are trained from scratch [9], [10].

To address the aforementioned issues, this paper proposes an improved YOLO-based model, termed YOLO-Rail, for railway track obstacle detection. The model incorporates ConvNeXtV2 [11] modules and depthwise separable convolutions into the network architecture to enhance feature extraction capabilities and reduce computational complexity. Furthermore, the feature fusion network is improved by integrating PAN and FPN. Meanwhile, a transfer learning strategy is adopted, where the model is first pre-trained on the large-scale COCO dataset [12] and then fine-tuned using the pre-trained weights on a railway track obstacle dataset. This approach effectively enhances both the detection performance and generalization ability of the model.

The main contributions of this paper are as follows:

- 1) An improved YOLO-Rail model is proposed to address the characteristics of railway track scenes, which enhances both detection accuracy and real-time performance.
- 2) The ConvNeXtV2 module and depthwise separable con-

This work was supported by the School of Electrical Engineering, Beijing Jiaotong University. (Corresponding author: Xin Du.)

Zhanyang Wang is pursuing a master's degree at the School of Electrical Engineering, Beijing Jiaotong University, Beijing 100044, China (e-mail: 23126367@bjtu.edu.cn).

Xin Du is with the School of Electrical Engineering, Beijing Jiaotong University, Beijing 100044, China (e-mail: xdu@bjtu.edu.cn).

volution are introduced to strengthen the feature representation capability of the model, thereby improving detection accuracy.

- 3) A feature fusion network is designed based on PAN and FPN architectures, adopting a top-down $\rightarrow$ bottom-up $\rightarrow$ top-down fusion strategy. This design enables efficient integration of multi-scale features and enhances the detection performance for small, medium, and large railway obstacles.
- 4) Through pre-training on the COCO dataset and fine-tuning on a railway obstacle dataset, the model's robustness and generalization performance in small-sample specific scenarios are significantly enhanced.
- 5) In experiments on railway obstacle detection, YOLO-Rail achieves superior detection results compared to mainstream object detection algorithms, demonstrating the effectiveness and application potential of the proposed model.

The remainder of this paper is organized as follows: Section II reviews related work; Section III describes the YOLO-Rail architecture and its main improvements; Section IV presents the experimental setup and performance evaluation; and Section V concludes the paper and discusses future research directions.

II. RELATED WORK

Object detection is one of the core tasks in the field of computer vision, primarily including two-stage approaches based on region proposals and end-to-end one-stage methods. Two-stage methods, represented by the R-CNN series [13]–[15], achieve high detection accuracy through region proposal extraction followed by classification and regression, but their inference speed is relatively slow, making it difficult to meet real-time requirements. One-stage methods, exemplified by YOLO [4] and SSD [16], perform bounding box regression and category prediction directly on feature maps, balancing both speed and accuracy. As a result, they have been widely adopted in real-time detection tasks. In recent years, with advancements in deep convolutional networks and visual Transformers, object detection methods have continued to improve in terms of accuracy, robustness, and capability in detecting small objects. In addition to CNNs and Transformers, Graph Neural Networks (GNNs) and their variants (such as GCN and DCAGGCN) [17]–[19] have emerged as powerful tools for deep feature extraction in recent years. By modeling the topological structure of data, these methods can effectively capture long-range dependencies and complex spatial relationships.

Since its introduction, the YOLO series has been continuously updated and has gradually become a mainstream method for real-time object detection. YOLOv3 [20] introduced multi-scale feature fusion to improve small object detection capabilities; YOLOv4 [21] incorporated CSPDarkNet and PANet structures, enhancing feature extraction efficiency and detection accuracy; YOLOv5 [22] excelled in engineering and lightweight design, facilitating deployment on embedded devices; YOLOv7 [23] and YOLOv8 [24] further optimized

the detection head and training strategies, achieving a new balance between speed and accuracy. Meanwhile, researchers have also attempted to integrate modules such as ConvNeXt [25], Transformer [26], and lightweight convolutions into the YOLO series models [27]–[29] to enhance their feature representation capabilities and cross-domain adaptability.

In the field of traffic safety, object detection techniques have been applied to tasks such as vehicle detection, pedestrian detection, traffic sign recognition, and railway scene monitoring [30]–[35]. For railway systems, some studies have attempted to use traditional computer vision methods (e.g., edge detection, background modeling) for obstacle identification, but their robustness remains limited [36]–[38]. In recent years, deep learning approaches have been gradually introduced to railway track monitoring tasks, such as foreign object detection and defect recognition based on Faster R-CNN or YOLO architectures [9], [39], with certain achievements. However, these methods still face two major challenges: first, the scarcity and diversity of railway obstacle data restrict the training and generalization capabilities of the models; second, the real-time performance and small object detection accuracy of existing models in complex scenarios need further improvement.

Overall, while existing research has made significant progress in object detection and traffic safety, the detection of obstacles on railway tracks still faces challenges such as insufficient data, limited detection accuracy, and constrained real-time performance [40], [41]. To address these issues, this paper proposes an improved YOLO-based model named YOLO-Rail. By incorporating the ConvNeXtV2 module and depthwise separable convolutions, the model enhances its feature extraction capabilities. Furthermore, a transfer learning strategy is adopted to improve generalization in small-sample scenarios, ultimately achieving superior detection performance for railway track obstacle detection tasks. To further highlight the differences between our approach and existing railway obstacle detection methods, a comparative analysis with representative studies is summarized in Table I.

III. METHOD

The YOLO-Rail model proposed in this paper builds upon the YOLO series and introduces improvements in both network architecture and training strategies for the task of railway track obstacle detection. The overall structure of the model, as shown in Fig. 1, consists of three main components: Backbone, Neck, and Head.

A. Backbone: Feature Extraction Network

The feature extraction network is primarily constructed by stacking CBL modules and an improved C2f module. Starting from the input image, the network gradually reduces the feature map size through convolution and downsampling operations while increasing the number of channels to enhance semantic information. Specifically, the input image first passes through several layers of CBL and improved C2f modules, outputting feature maps at different scales. The feature maps at each layer contain multi-scale features ranging from low-level texture information to high-level semantic information.

TABLE I: Comparison with Existing Railway Obstacle Detection Methods

Reference	Year	Method / Key Technique	Dataset	Main Characteristics
Xu et al. [31]	2019	Real-time CNN-based obstacle detector for rail environments	Authors' private railway rail-view dataset	Demonstrated feasibility of deep CNNs for real-time rail obstacle detection
Wang et al. [42]	2021	Semi-supervised autoencoder-based foreign object detection	Private railway monitoring images	Effective under limited labeled anomaly samples
Rahman et al. [35]	2022	Lightweight DNN with transfer learning (MobileNetV2 backbone)	Self-collected railway obstacle images	Designed for real-time and embedded deployment scenarios
Chen et al. [43]	2024	RailFOD23 dataset and baseline detector benchmarking	RailFOD23 public dataset	Provides standardized large-scale dataset and baseline evaluations
Niu and Hou [9]	2025	MSL-YOLO: Multi-scale lightweight YOLO architecture	Custom railway foreign-object dataset	Improves detection accuracy with reduced model complexity
Guo et al. [10]	2025	Synthetic data generation for railway obstacle detection	SynRailObs synthetic(train) + real data(test)	Enhances detection robustness for rare railway obstacles

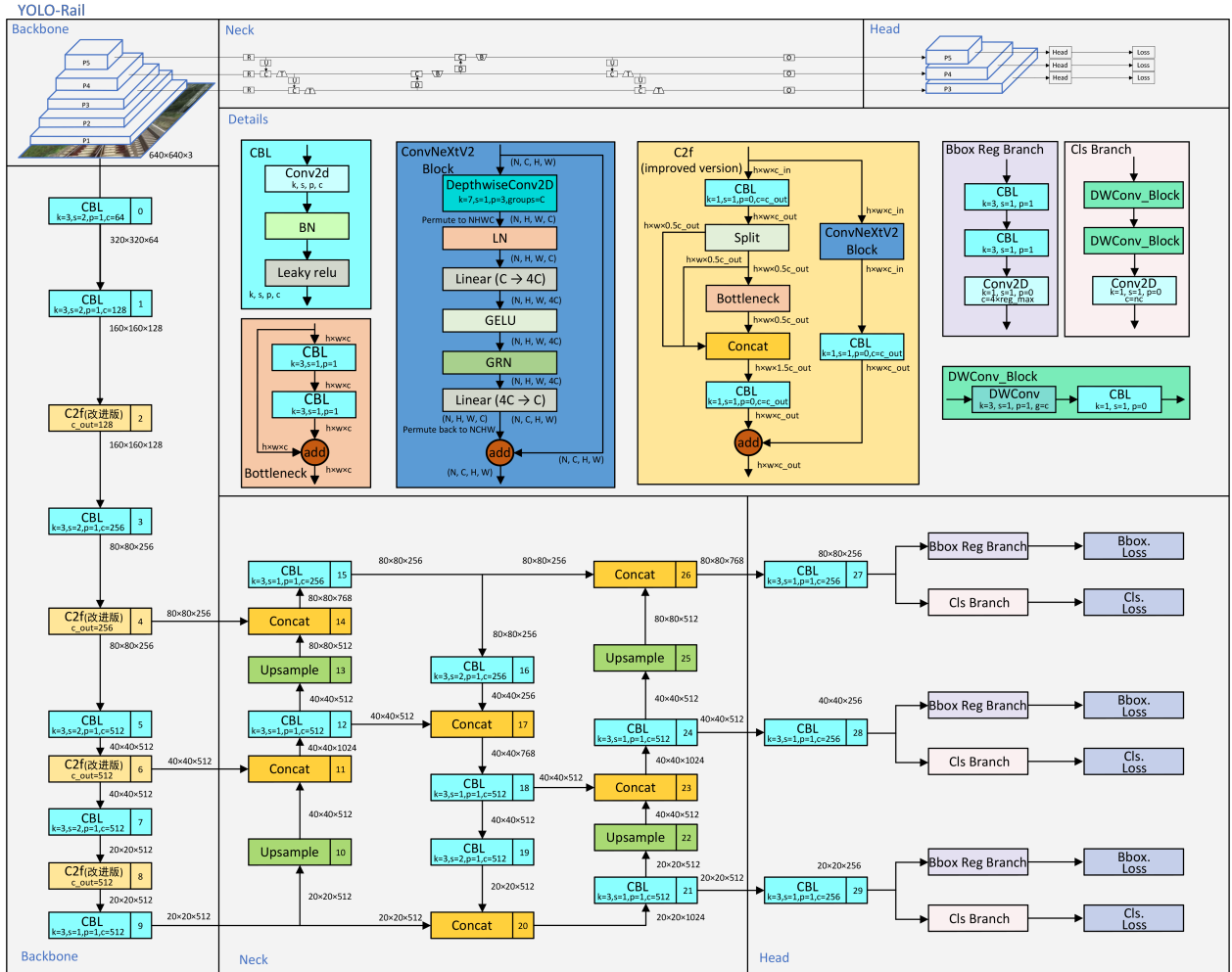


Fig. 1: Schematic Diagram of the Overall YOLO-Rail Architecture

Compared to the original YOLO backbone, YOLO-Rail incorporates the ConvNeXTv2 Block to enhance the C2f module, providing the network with stronger representational capacity when extracting deep features. Unlike generic natural-image datasets, railway scenes exhibit strong geometric regularity and long-range spatial continuity along the track direction. The introduction of ConvNeXTv2 blocks enables more effective modeling of such global structural patterns, which is beneficial for suppressing background clutter and improving obstacle-background discrimination in rail environments. This design

ensures that the backbone retains detailed shallow features while improving the semantic understanding of deep features. Through these improvements, the backbone can extract more robust low-level and high-level semantic features in complex railway track scenarios. Figure 2 illustrates the original C2f module [24] and the improved version. Compared to the original structure, the improved module not only retains efficient residual connection mechanisms but also introduces global modeling capabilities, making it better suited for object detection tasks in complex backgrounds.

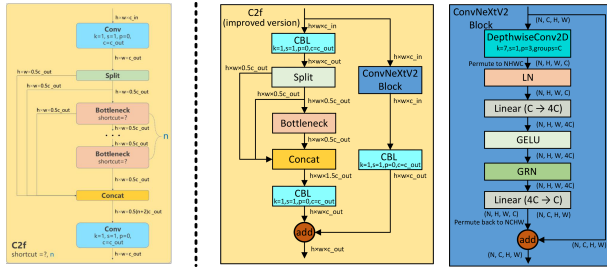


Fig. 2: C2f (left) and improved C2f (right)

B. Neck: Feature Fusion Network

The feature fusion network primarily draws inspiration from PAN (Path Aggregation Network) [44] and FPN (Feature Pyramid Network) [45], incorporating a top-down path → bottom-up path → top-down fusion process. First, a top-down path is adopted to progressively upsample high-level semantic information and concatenate it with shallow features, thereby enhancing the detailed representation of small objects. Subsequently, a bottom-up path is used to downsample the fused features and integrate them with deep features, ensuring semantic completeness for large objects. This is followed by another top-down fusion step. Ultimately, the network outputs feature maps at three scales: 20×20 , 40×40 , and 80×80 , corresponding to the detection requirements for large, medium, and small objects, respectively. This architecture enhances multi-scale feature integration and improves detection performance.

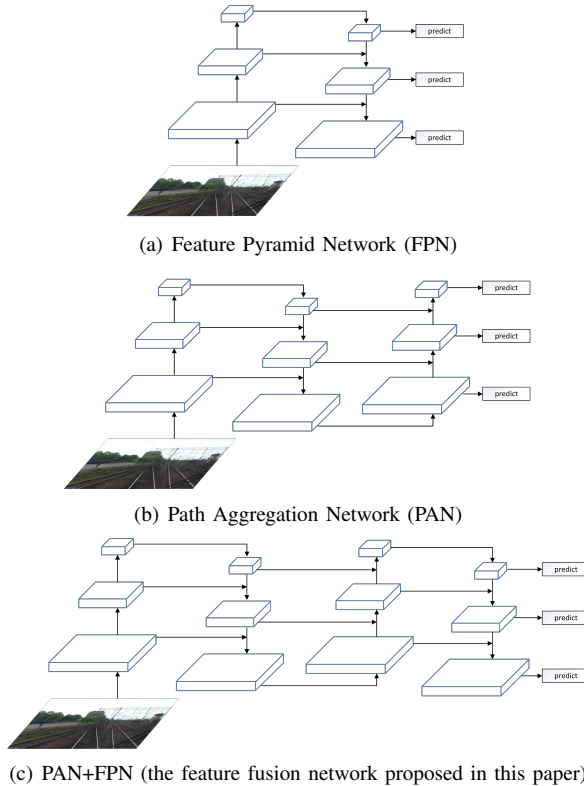


Fig. 3: Design of Feature Fusion Networks

C. Head: Detection Head

In the detection head, the model adopts a dual-branch structure to separately model bounding box regression and object classification. The bounding box regression branch employs two standard convolutional layers followed by a 1×1 convolutional layer to generate a distributed regression representation with $4 \times \text{reg_max}$ channels, enabling high-precision bounding box prediction. The classification branch incorporates depthwise separable convolution [46], utilizing a two-stage stack of DWConv + 1×1 Conv to reduce computational complexity and parameter count while enhancing inter-channel feature interaction. Finally, a 1×1 convolution outputs the category prediction results. The outputs of the two branches are fused along the channel dimension to form a prediction tensor containing both bounding box and category information. The bounding box component is decoded into continuous coordinates via Distributed Focal Loss (DFL), and the classification component is processed through a sigmoid function to obtain class probabilities, achieving efficient and accurate object detection.

D. Loss Function

The loss function in this object detection task is typically composed of multiple components, jointly optimizing the performance of both bounding box regression and object classification. The overall loss function is defined as

$$L = \lambda_{box} L_{box} + \lambda_{cls} L_{cls} + \lambda_{dfl} L_{dfl}, \quad (1)$$

where L_{box} denotes the bounding box regression loss, L_{cls} the classification loss, and L_{dfl} the distribution focal loss. The coefficients λ_{box} , λ_{cls} , and λ_{dfl} are the corresponding weighting factors.

a) *Bounding Box Regression Loss*: The bounding box regression loss measures the positional discrepancy between predicted and ground-truth boxes. A common form is based on the Intersection over Union (IoU) metric. Given a predicted box B_p and a ground-truth box B_g , IoU is defined as

$$\text{IoU}(B_p, B_g) = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}. \quad (2)$$

Based on this, the Generalized IoU (GIoU) loss is adopted:

$$L_{box} = 1 - \text{IoU}(B_p, B_g) + \frac{|C - (B_p \cup B_g)|}{|C|}, \quad (3)$$

where C represents the smallest enclosing box covering both B_p and B_g . This loss not only accounts for the overlapping region but also provides effective gradients when the boxes do not intersect.

b) *Classification Loss*: The classification loss quantifies the discrepancy between the predicted class probability distribution $\hat{p}$ and the ground-truth label y . To address the imbalance between positive and negative samples, a weighted binary cross-entropy form of Focal Loss is employed:

$$L_{cls} = - \sum_{c=1}^C \alpha (1 - \hat{p}_c)^\gamma y_c \log(\hat{p}_c), \quad (4)$$

where C is the total number of classes, α is a balancing factor, and γ is the focusing parameter. When $\gamma > 0$, this loss reduces the weight of easily classified samples and emphasizes hard examples during training.

c) *Distribution Focal Loss*: To enhance the precision of bounding box coordinate prediction, Distribution Focal Loss (DFL) is utilized. The core idea is to transform the continuous coordinate regression problem into discrete distribution modeling. Specifically, each bounding box coordinate is discretized into K bins, and the model outputs a probability distribution $\hat{q} \in \mathbb{R}^K$ over these bins. The ground-truth coordinate t is represented by a soft label vector $q \in \mathbb{R}^K$, distributed over adjacent bins. The loss function is defined as

$$L_{df} = - \sum_{k=1}^K q_k \log(\hat{q}_k), \quad (5)$$

which measures the cross-entropy between the predicted and target distributions. The final continuous prediction is recovered by the weighted sum over all bins:

$$\hat{t} = \sum_{k=1}^K k \cdot \hat{q}_k. \quad (6)$$

This approach captures fine-grained probabilistic information of bounding box coordinates, leading to more accurate localization compared to direct regression, particularly effective in small object detection scenarios.

In summary, this composite loss function jointly optimizes localization, classification, and coordinate distribution modeling, thereby improving the overall detection performance through integrated constraints on both positioning and recognition.

IV. EXPERIMENT

A. Dataset

The rail obstacle detection dataset comprises a total of 3,911 images covering eight critical categories: animal, bicycle, bus-truck, car, other-vehicles, person, train, and turnout.detection. All images in the rail obstacle detection dataset are originally captured at a resolution of 640×640 pixels, which directly matches the input size required by the YOLO-based detection framework. The annotations follow the standard YOLO format, where each object is represented by a class label along with normalized bounding box coordinates (x, y, w, h) , denoting the center location and width–height of the bounding box. The dataset was collected under real-world railway monitoring conditions with diverse and complex environments, including urban areas, mountainous regions, and tunnel scenarios. In addition, it covers a wide range of illumination and weather conditions, such as daytime, nighttime, and snowy scenes, reflecting realistic temperature and environmental variations encountered in railway operations. Due to practical deployment constraints in railway systems, the camera mounting height is not fixed and varies across different acquisition locations. This variability introduces multiple viewpoints and scales in the dataset, increasing its overall complexity and enabling the trained model to better generalize to real-world railway obstacle detection tasks.

The detailed distribution of instances across the eight categories is presented in Table II. As observed, the dataset

TABLE II: Distribution of Instances per Category in the Railway Obstacle Detection Dataset

Category	Instances	Proportion (%)
car	9,211	49.85
person	5,615	30.39
train	1,596	8.64
bus-truck	897	4.85
bicycle	707	3.83
turnout.detection	248	1.34
other-vehicles	186	1.01
animal	19	0.10
Total	18,479	100.00^a

^a Note: Percentages may not total 100% due to rounding.

exhibits a long-tailed distribution, which is consistent with the real-world frequency of objects in railway environments. Common objects like car and person appear frequently, while safety-critical anomalies like animal or other-vehicles are naturally rare. Although this class imbalance poses a challenge, we mitigate its impact through two strategies: 1) Transfer Learning: As mentioned in the pre-training stage, the model is pre-trained on the COCO dataset. Since COCO contains rich samples of animals and vehicles, the model acquires prior feature representations for these minority classes before fine-tuning. 2) Data Augmentation: Strategies such as Mosaic (detailed in APPENDIX I) increase the effective occurrence of rare objects within training batches by stitching images together. A schematic illustration of the dataset is shown in Fig. 4. The dataset is randomly split into training and



Fig. 4: Schematic Diagram of the Rail Obstacle Detection Dataset

validation sets in a 7:3 ratio, where the training set is used for model parameter optimization and the validation set is used

to evaluate the model's performance. Although the dataset follows a 7:3 split ratio, the training and validation sets are manually fixed by organizing images into predefined train and val folders, rather than being randomly re-split at each run.

B. Training Setup

During the model training phase, this study employs a GPU workstation equipped with four NVIDIA RTX 3090 (24GB VRAM) GPUs for parallel training. The training pipeline is implemented based on the PyTorch 1.11 deep learning framework and Python 3.9 environment. In addition, to ensure reproducibility, the random seed for all experiments is fixed to 0, including data loading, data augmentation, and model initialization. The training process consists of two stages: pre-training and fine-tuning. In the pre-training stage, the YOLO-rail model is initially trained on the large-scale public dataset COCO2017 to enhance its feature extraction and generalization capabilities by leveraging the dataset's rich category and scene prior knowledge. The model's performance is preliminarily evaluated using the COCO validation set. Upon completion of this stage, a pre-trained weight with general object detection capabilities is obtained. Subsequently, in the fine-tuning stage, the pre-trained weights are loaded into the model, and targeted training is conducted based on a rail obstacle detection dataset. Through the transfer learning strategy, the model effectively adapts to the obstacle detection task in rail scenarios, improving both training efficiency and enhancing recognition performance and robustness in the target domain.

C. Ablation Study

The ablation study results are summarized in Table III to quantitatively analyze the individual contributions of the key components in YOLO-Rail-s. Here, w/o denotes the variant where a specific module is removed while all other settings remain unchanged, and Baseline refers to the full YOLO-Rail-s model.

TABLE III: Ablation study on key components of YOLO-Rail-s(from scratch).

Model Variant	mAP	mAP @0.50	mAP @s	AR @s	Params (M)	GFLOPs
Baseline	17.1	31.7	9.2	23.3	12.1	47.8
w/o ConvNeXtV2	16.4	30.3	9.5	23.9	10.5	41.3
w/o DWConv	16.9	31.4	9.9	24.6	11.8	47.0
w/o FPN (PAN only)	16.8	31.4	9.6	24.3	10.3	39.5

Removing the ConvNeXtV2 module results in a noticeable performance degradation, with the overall mAP dropping from 17.1 to 16.4. This indicates that the ConvNeXtV2-based feature extraction module effectively enhances the representation capability of the backbone, especially in complex railway scenes. When the depthwise separable convolution (DWConv) is replaced with standard convolution in the detection head, the mAP decreases to 16.9, indicating that DWConv contributes to more effective feature representation in the detection head. In addition, removing the FPN top-down pathway and retaining

only PAN leads to a decline in overall accuracy performance, suggesting that the proposed multi-scale feature fusion design is beneficial for capturing fine-grained obstacle features.

It is worth noting that, although the removal of certain components leads to a slight improvement in small-object detection performance (mAP@s or AR@s), the overall detection accuracy consistently degrades. This phenomenon suggests a trade-off between global feature representation and local feature sensitivity. While simplified architectures may favor small-scale objects by emphasizing local texture information, they lack sufficient semantic context to maintain robust performance across objects of all scales. In contrast, the full YOLO-Rail-s model achieves a more balanced performance by integrating multi-scale feature fusion and enhanced backbone representation, resulting in superior overall detection accuracy.

Overall, the ablation results confirm that each component contributes positively to the final performance, and their combination in the full YOLO-Rail-s configuration achieves the best balance between accuracy and efficiency.

D. Main Results

The FPS is obtained with a batch size of 1 using mixed-precision inference (FP16). The models are trained for 300 epochs with an early stopping strategy set to a patience of 100. In the experimental results table, mAP@0.50:0.95 is denoted as mAP, and GFLOPs are calculated based on an input image resolution of 640×640 . The metrics mAP, mAP@0.50, mAP@s, and AR@s are expressed as percentages (%).

As shown in Table IV, a comparison of the detection performance between the anchor-based and anchor-free detection heads of YOLO-Rail on the rail obstacle dataset is presented.

TABLE IV: Comparison of different detection head strategies for YOLO-Rail-s trained from scratch on the rail obstacle detection dataset.

Detection Head	mAP	mAP @0.50	mAP @s	AR @s	FPS	Params (M)	GFLOPs
Anchor-based	15.6	31.0	10.1	26.9	126.42	10.5	40.0
Anchor-free	17.1	31.7	9.2	23.3	96.46	12.1	47.8

Table IV presents a comparative analysis of different detection head strategies. In terms of overall detection accuracy on the validation set, the Anchor-free strategy achieves an mAP of 17.1, which is significantly higher than the 15.6 obtained by the Anchor-based approach. The mAP@0.5 also improves from 31.0 to 31.7, indicating that the Anchor-free strategy offers superior overall performance in object detection. Although the Anchor-based method slightly outperforms the Anchor-free method in small object detection metrics, with mAP@s and AR@s scores of 10.1 and 26.9 compared to 9.2 and 23.3, respectively, the difference is not substantial. On the other hand, in terms of computational cost, the Anchor-based approach has fewer parameters (10.5M) and lower computational complexity (40.0 GFLOPs) than the Anchor-free method (12.1M and 47.8 GFLOPs), and it also demonstrates a slightly faster inference speed (126.42 FPS vs. 96.46 FPS). Both approaches, however, meet real-time requirements. Overall, the

Anchor-based method holds a slight advantage in efficiency and small object recall, while the Anchor-free method excels in overall accuracy.

It is worth noting that although the anchor-based detection head exhibits slightly higher mAP@s and AR@s scores, the overall performance difference in small object detection is relatively limited. More importantly, from the perspective of practical railway applications, anchor-free detection presents several advantages that are particularly relevant to safety-critical scenarios. In real railway environments, small obstacles such as animals, fallen debris, or foreign objects often appear at varying distances and exhibit highly diverse scales and aspect ratios. Anchor-based methods rely on predefined anchor sizes, which require careful manual design and tuning. In real-world deployment, mismatches between anchor configurations and unseen obstacle scales may lead to missed detections, especially for rare or safety-critical small targets. In contrast, anchor-free detection eliminates the dependency on anchor priors and directly regresses object locations, enabling more flexible adaptation to continuously varying object scales.

Moreover, the anchor-free detection head demonstrates superior overall detection accuracy, which is essential for maintaining robust perception performance in complex railway scenes. When combined with the proposed multi-scale feature fusion strategy and transfer learning scheme, the anchor-free design maintains competitive small object detection capability while offering improved generalization and stability. Therefore, considering both detection accuracy and practical reliability in real railway systems, the anchor-free detection head is selected as a more suitable choice for YOLO-Rail.

The pre-training results of the anchor-free detection head model on the large-scale COCO dataset are shown in Table V. The fine-tuning results of the pre-trained model on the

TABLE V: Pre-training Results on the Large-Scale COCO Dataset

Model	mAP	mAP @0.50	mAP @s	AR @s	FPS	Params (M)	GFLOPs
YOLO-Rail-n	28.5	42.0	14.6	32.5	105.08	3.2	13.4
YOLO-Rail-s	34.4	49.4	18.6	37.5	96.46	12.1	47.8
YOLO-Rail-m	38.4	54.1	21.6	42.3	91.34	26.1	103.7
YOLO-Rail-l	42.4	59.7	28.3	46.6	70.61	46.2	182.5
YOLO-Rail-x	44.8	61.7	27.9	44.6	48.76	72.0	284.2

rail obstacle detection dataset are shown in Table VI.

TABLE VI: Fine-tuning results based on pre-trained models.

Model	mAP	mAP @0.50	mAP @s	AR @s	FPS	Params (M)	GFLOPs
YOLO-Rail-n	16.0	29.0	7.5	24.6	105.08	3.2	13.4
YOLO-Rail-s	18.5	32.7	10.5	24.6	96.46	12.1	47.8
YOLO-Rail-m	20.0	33.6	10.5	27.6	91.34	26.1	103.7
YOLO-Rail-l	21.3	36.4	12.1	29.8	70.61	46.2	182.5
YOLO-Rail-x	24.7	40.6	12.5	30.6	48.76	72.0	284.2

Table VII compares the performance of YOLO-Rail-s trained from scratch and with COCO pre-training on the rail obstacle detection dataset. The results show that COCO pre-training consistently improves detection accuracy, with mAP increasing from 17.1% to 18.5% and mAP@0.50 from 31.7% to 32.7%. Improvements are also observed for small object detection, as indicated by higher mAP@s and AR@s.

These results confirm that pre-training on a large-scale generic dataset provides more robust feature representations and effectively enhances performance in data-limited railway scenarios.

TABLE VII: Effect of COCO pre-training on rail obstacle detection performance using YOLO-Rail-s.

Training strategy	Pre-training	mAP	mAP@0.50	mAP@s	AR@s
From scratch	✗	17.1	31.7	9.2	23.3
Transfer learning	✓ (COCO)	18.5	32.7	10.5	24.6

The performance of mainstream object detection algorithms, including YOLOv5, YOLOv11 [46], YOLOv12 [47], and YOLOX [48], on the rail obstacle dataset is presented in Table VIII.

TABLE VIII: Comparison of the Effectiveness of Various Object Detection Algorithms.

Model	mAP	mAP @0.50	mAP @s	AR @s	FPS	Params (M)	GFLOPs
YOLOv5-n	11.1	25.1	5.4	22.3	133.90	1.8	4.2
YOLOv5-s	14.6	30.0	8.3	23.2	128.53	7.2	16.4
YOLOv5-m	15.4	30.5	9.0	23.9	94.97	20.9	47.9
YOLOv5-l	16.1	30.9	9.0	23.6	73.10	46.1	107.8
YOLOv5-x	16.9	32.7	10.0	24.0	61.54	86.2	203.9
YOLOv11-n	13.7	26.3	6.4	22.4	93.59	2.6	6.4
YOLOv11-s	17.0	31.4	10.0	25.5	90.66	9.4	21.6
YOLOv11-m	17.8	32.7	10.1	24.5	69.76	20.1	68.2
YOLOv11-l	19.3	33.9	11.5	27.4	48.38	25.3	87.3
YOLOv11-x	19.4	34.6	11.9	26.9	44.7	56.9	195.5
YOLOv12-n	13.1	25.2	6.4	22.1	68.29	2.6	6.5
YOLOv12-s	16.7	30.3	9.5	25.4	66.47	9.3	21.7
YOLOv12-m	18.1	33.2	11.5	27.9	55.43	20.1	67.8
YOLOv12-l	19.2	34.3	11.4	26.9	36.47	26.4	89.4
YOLOv12-x	19.2	34.1	11.6	26.5	33.97	59.1	199.9
YOLOX-s	16.3	32.7	9.9	28.7	38.7	8.9	13.3
YOLOX-m	18.6	36.3	12.9	29.2	30.4	25.3	36.8
YOLOX-l	19.1	37.0	12.8	28.8	26.1	54.2	77.7
YOLOX-x	19.3	37.4	12.8	29.9	21.8	99.0	141.0

Experimental results demonstrate that the YOLO-Rail series models outperform mainstream detection algorithms in railway scene detection tasks (Table VIII). On the one hand, as the model scale increases, the detection accuracy of YOLO-Rail improves progressively, with YOLO-Rail-x achieving 24.7 mAP and 40.6 mAP@0.5, significantly surpassing other lightweight models. On the other hand, YOLO-Rail exhibits outstanding performance in both small object detection and recall rate, indicating its stronger suitability for fine-grained object recognition in complex scenarios. Compared to models such as YOLOv5, YOLOv11, YOLOv12, and YOLOX, YOLO-Rail generally achieves higher detection accuracy under comparable model sizes, while its lightweight versions (YOLO-Rail-n/s) maintain a real-time inference speed of over 90 FPS, striking a favorable balance between detection accuracy and inference efficiency. The superior performance of YOLO-Rail on railway obstacle detection validates that its architectural improvements are exceptionally well-suited to such scenarios.

It is worth noting that the absolute mAP values reported in Table VIII are notably lower than commonly reported YOLO benchmarks on generic datasets such as COCO. This

discrepancy is primarily attributed to the inherent difficulty of the railway obstacle detection task rather than limitations of the proposed model. Specifically, the dataset is dominated by small-scale objects captured under long viewing distances, complex backgrounds, and frequent occlusions, which significantly increase localization and classification difficulty. In addition, the dataset exhibits a strong long-tailed class distribution, where rare but safety-critical categories (e.g., animal, turnout detection, and other-vehicles) are severely underrepresented. These factors jointly constrain the achievable upper bound of mAP under standard evaluation protocols.

Importantly, all compared models are trained and evaluated under identical conditions on the same dataset. Therefore, the consistent performance gains achieved by YOLO-Rail across different model scales demonstrate its superior robustness and effectiveness in handling small, sparse, and fine-grained obstacles in real-world railway scenarios.

V. CONCLUSION

To meet the specific requirements of object detection in railway scenarios, this paper proposes an improved YOLO-Rail model. The backbone incorporates lightweight ConvNeXtV2 modules and depthwise separable convolutions to balance feature extraction capability and computational efficiency. The feature fusion network combines PAN and FPN structures, including two top-down paths and one bottom-up path, to better integrate multi-scale features. In the detection head, a decoupled design is adopted to effectively enhance the collaborative optimization of bounding box regression and classification tasks.

Experimental results demonstrate that YOLO-Rail outperforms mainstream models including YOLOv5, YOLOv11, YOLOv12, and YOLOX across various scales. In terms of accuracy, YOLO-Rail-x achieves an mAP of 24.7 and an mAP@0.5 of 40.6, significantly surpassing models of comparable size. It also exhibits superior performance in small object detection and recall rate, validating its applicability in complex railway scenarios. Meanwhile, the lightweight versions, YOLO-Rail-n and YOLO-Rail-s, maintain high detection accuracy while achieving inference speeds exceeding 90 FPS, meeting the requirements for real-time detection.

In summary, the YOLO-Rail model not only outperforms existing mainstream algorithms in detection accuracy, but also achieves a favorable balance between object detection capability and real-time performance, demonstrating strong potential for practical deployment in railway scene object detection. Nevertheless, there remains room for further optimization of YOLO-Rail. Future research may explore the following directions: first, investigating Transformer-based feature modeling methods to further enhance the recognition capability for complex backgrounds and long-range dependent objects; second, incorporating multimodal data (such as infrared, radar, or trackside sensor information) to improve the model's robustness under extreme lighting and adverse weather conditions; third, conducting more in-depth model compression and acceleration optimization for edge computing devices to enhance deployment efficiency in practical railway

monitoring systems. Through explorations in these directions, YOLO-Rail is expected to play a greater role in a wider range of railway safety and intelligent monitoring scenarios.

APPENDIX I HYPERPARAMETER SETTINGS

To ensure stable optimization and fair comparison with existing YOLO-based detectors, the hyperparameters used in both the pre-training and fine-tuning stages are carefully selected based on widely adopted configurations in prior works and empirical validation. Most optimization-related parameters, including the SGD optimizer, momentum, and weight decay, follow standard YOLO settings that have been proven effective for large-scale object detection tasks.

Specifically, an initial learning rate of 0.01 is adopted during pre-training on the COCO dataset to facilitate fast convergence on large-scale data, while a smaller learning rate of 0.001 is used during fine-tuning to ensure stable adaptation to the relatively small railway obstacle dataset and to mitigate overfitting. A linear learning rate decay schedule is employed to gradually reduce the learning rate, improving convergence stability in later training stages. The loss gain coefficients for bounding box regression, classification, and distribution focal loss are inherited from commonly used YOLO configurations and empirically verified to maintain a balanced contribution among localization accuracy, classification confidence, and fine-grained bounding box regression. Regarding data augmentation, color-space augmentations in the HSV domain are applied to enhance robustness under varying illumination conditions commonly encountered in railway scenes. Geometric augmentations such as translation, scaling, mosaic, and horizontal flipping are employed to increase sample diversity and improve generalization capability, particularly for rare obstacle categories. At the same time, augmentation intensities are moderately constrained to avoid disrupting the geometric structure of railway track scenes.

The complete hyperparameter settings are summarized in Table IX.

TABLE IX: Hyperparameter Settings

Parameters	Value
Optimizer	SGD
Lr0 (Pre-training/Fine-tuning)	0.01/0.001
Lrf	0.01
Momentum	0.937
Weight_decay	0.0005
Warmup_epochs	3.0
Warmup_momentum	0.8
Warmup_bias_lr	0.1
Box loss gain	7.5
Class loss gain	0.5
DFL loss gain	1.5
HSV hue augmentation	0.4
HSV saturation augmentation	0.015
HSV value augmentation	0.7
Learning rate schedule	Linear decay
Translation augmentation	0.1
Scale augmentation	0.5
Mosaic augmentation	1.0
Copy_paste_mode	flip
Fliplr	0.5

APPENDIX II

VISUALIZED RESULTS

To further demonstrate the effectiveness of the proposed YOLO-Rail-x model, several representative prediction results on the rail obstacle dataset are visualized in Fig. 5. The visualized results clearly show that the model is capable of accurately detecting and locating various types of obstacles. The bounding boxes generated by the model align well with the ground-truth annotations, indicating strong localization ability and robustness to environmental noise.

Fig. 6 further presents representative failure cases of the YOLO-Rail model in challenging railway scenarios. In the first image, a missed detection of a train is observed in a nighttime scenario. Although the train is perceptible to human observers, strong headlight illumination and multiple surrounding light sources significantly reduce the visibility of structural cues. Such illumination artifacts and viewpoint bias can weaken the discriminative features learned by the model, leading to false negatives in challenging low-light environments. In the second image, missed detections of trains are observed in nighttime scenes. Under extremely low illumination, the visual contrast between the train and the background is significantly reduced, and image noise becomes more prominent, which degrades the robustness of feature extraction. In the last two images, missed detections of small pedestrians occur in daytime scenes. Although lighting conditions are favorable, these targets occupy only a very limited number of pixels and often appear in complex backgrounds, making accurate detection difficult.

These representative failure cases highlight that extreme low-light environments and small-scale targets remain challenging scenarios in practical railway obstacle detection. Despite the overall strong performance of YOLO-Rail, the degradation of discriminative visual cues under such conditions may limit the reliability of purely vision-based detection.



Fig. 5: Visualized prediction results of the YOLO-Rail-x model on the rail obstacle dataset

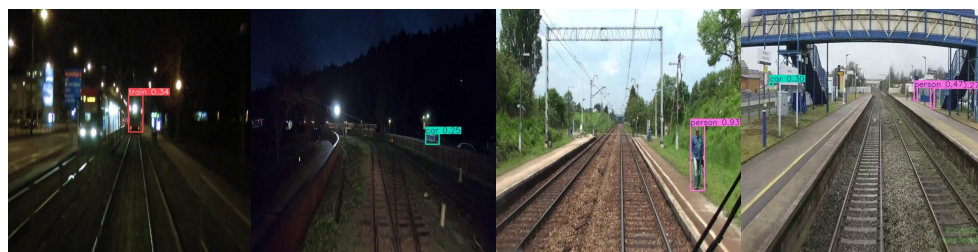


Fig. 6: Failure cases of the YOLO-Rail-x model in challenging railway scenarios.

REFERENCES

- [1] X. Tianwen, X. Yongneng, and Y. Huimin, "Research on obstacle detection method of urban rail transit based on multisensor technology," *Journal of Artificial Intelligence and Technology*, vol. 1, no. 1, pp. 61–67, 2021.
- [2] R. RS, T. Al-Shehari, S. Nathan, J. A. S. R. S. P. P. T. Alfakih, and H. Alsaman, "An unmanned aerial vehicle captured dataset for railroad segmentation and obstacle detection," *Scientific Data*, vol. 11, no. 1, p. 1315, 2024.
- [3] F. Hampel, P. Leibner, H. Manjunatheswaran, A. Schotten, and C. Schindler, "Obstacle detection on rail-tracks: Study on situations for requirement specification," *Transportation research procedia*, vol. 72, pp. 884–891, 2023.
- [4] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 779–788, 2016.
- [5] X. Chen, H. Ma, J. Wan, B. Li, and T. Xia, "Multi-view 3d object detection network for autonomous driving," in *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition*, pp. 1907–1915, 2017.
- [6] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in *2017 IEEE international conference on image processing (ICIP)*, pp. 3645–3649, IEEE, 2017.
- [7] L. T. Ramos and A. D. Sappa, "A decade of you only look once (yolo) for object detection," *arXiv preprint arXiv:2504.18586*, 2025.
- [8] M. Kotthapalli, D. Ravipati, and R. Bhatia, "Yolov1 to yolov11: A comprehensive survey of real-time object detection innovations and challenges," *arXiv preprint arXiv:2508.02067*, 2025.
- [9] H. Niu, D. Feng, and T. Hou, "Research on foreign object intrusion detection in railway tracks based on msl-yolo," *Journal of Engineering and Applied Science*, vol. 72, no. 1, p. 136, 2025.
- [10] Q. Guo and J. Rambach, "Synrailobs: A synthetic dataset for obstacle detection in railway scenarios," *arXiv preprint arXiv:2505.10784*, 2025.
- [11] S. Woo, S. Debnath, R. Hu, X. Chen, Z. Liu, I. S. Kweon, and S. Xie, "Convnext v2: Co-designing and scaling convnets with masked autoencoders," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 16133–16142, 2023.
- [12] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft coco: Common objects in context," in *European conference on computer vision*, pp. 740–755, Springer, 2014.
- [13] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 580–587, 2014.
- [14] R. Girshick, "Fast r-cnn," in *Proceedings of the IEEE international conference on computer vision*, pp. 1440–1448, 2015.
- [15] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," *Advances in neural information processing systems*, vol. 28, 2015.
- [16] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "Ssd: Single shot multibox detector," in *European conference on computer vision*, pp. 21–37, Springer, 2016.
- [17] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The graph neural network model," *IEEE transactions on neural networks*, vol. 20, no. 1, pp. 61–80, 2008.
- [18] T. Kipf, "Semi-supervised classification with graph convolutional networks," *arXiv preprint arXiv:1609.02907*, 2016.
- [19] D. He, J. Zhao, Z. Jin, C. Huang, C. Yi, and J. Wu, "Dcagcn: A novel method for remaining useful life prediction of bearings," *Reliability Engineering & System Safety*, vol. 260, p. 110978, 2025.
- [20] J. Redmon and A. Farhadi, "Yolov3: An incremental improvement," *arXiv preprint arXiv:1804.02767*, 2018.
- [21] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "Yolov4: Optimal speed and accuracy of object detection," *arXiv preprint arXiv:2004.10934*, 2020.
- [22] G. Jocher, A. Stoken, J. Borovec, L. Changyu, A. Hogan, L. Diaconu, J. Poznanski, L. Yu, P. Rai, R. Ferriday, et al., "ultralytics/yolov5: v3.0," *Zenodo*, 2020.
- [23] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "Yolov7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 7464–7475, 2023.
- [24] R. Varghese and M. Sambath, "Yolov8: A novel object detection algorithm with enhanced performance and robustness," in *2024 International conference on advances in data engineering and intelligent computing systems (ADICS)*, pp. 1–6, IEEE, 2024.
- [25] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 11976–11986, 2022.
- [26] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, E. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.
- [27] F. Zhou, H. Deng, Q. Xu, and X. Lan, "Cntr-yolo: improved yolov5 based on convnext and transformer for aircraft detection in remote sensing images," *Electronics*, vol. 12, no. 12, p. 2671, 2023.
- [28] J. Zhou, B. Zhang, X. Yuan, C. Lian, L. Ji, Q. Zhang, and J. Yue, "Yolo-cir: The network based on yolo and convnext for infrared object detection," *Infrared Physics & Technology*, vol. 131, p. 104703, 2023.
- [29] J. Tang, X.-M. Hu, S.-W. Jeon, and W.-N. Chen, "Light-yolo: a lightweight detection algorithm based on multi-scale feature enhancement for infrared small ship target," *Complex & Intelligent Systems*, vol. 11, no. 2, p. 130, 2025.
- [30] Á. Arcos-García, J. A. Alvarez-García, and L. M. Soria-Morillo, "Evaluation of deep neural networks for traffic sign detection systems," *Neurocomputing*, vol. 316, pp. 332–344, 2018.
- [31] Y. Xu, C. Gao, L. Yuan, S. Tang, and G. Wei, "Real-time obstacle detection over rails using deep convolutional neural network," in *2019 IEEE Intelligent Transportation Systems Conference (ITSC)*, pp. 1007–1012, IEEE, 2019.
- [32] D. Tabernik and D. Skočaj, "Deep learning for large-scale traffic-sign detection and recognition," *IEEE transactions on intelligent transportation systems*, vol. 21, no. 4, pp. 1427–1440, 2019.
- [33] L. G. Galvao, M. Abbod, T. Kalganova, V. Palade, and M. N. Huda, "Pedestrian and vehicle detection in autonomous vehicle perception systems—a review," *Sensors*, vol. 21, no. 21, p. 7267, 2021.
- [34] Z.-J. Wan and S.-Q. Chen, "Railway tracks defects detection based on deep convolution neural networks," in *Artificial Intelligence in China: Proceedings of the 2nd International Conference on Artificial Intelligence in China*, pp. 119–129, Springer, 2021.
- [35] F. U. Rahman, M. T. Ahmed, M. M. Hasan, and N. Jahan, "Real-time obstacle detection over railway track using deep neural networks," *Procedia Computer Science*, vol. 215, pp. 289–298, 2022.
- [36] H. Niu and T. Hou, "Fast detection study of foreign object intrusion on railway track," *Archives of Transport*, vol. 47, no. 3, pp. 79–89, 2018.
- [37] Y. Min, B. Xiao, J. Dang, B. Yue, and T. Cheng, "Real time detection system for rail surface defects based on machine vision," *EURASIP Journal on Image and Video Processing*, vol. 2018, no. 1, pp. 1–11, 2018.
- [38] D. Ristić-Durrant, M. Franke, and K. Michels, "A review of vision-based on-board obstacle detection and distance estimation in railways," *Sensors*, vol. 21, no. 10, p. 3452, 2021.
- [39] T. Bai, J. Yang, G. Xu, and D. Yao, "An optimized railway fastener detection method based on modified faster r-cnn," *Measurement*, vol. 182, p. 109742, 2021.
- [40] M. Franke, V. Gopinath, D. Ristić-Durrant, and K. Michels, "Object-level data augmentation for deep learning-based obstacle detection in railways," *Applied Sciences*, vol. 12, no. 20, p. 10625, 2022.
- [41] X. CanYang, L. Yingying, L. Yongqiang, T. Runliang, and G. Tao, "Lightweight rail surface defect detection algorithm based on an improved yolov8," *Measurement*, vol. 242, p. 115922, 2025.
- [42] T. Wang, Z. Zhang, F. Yang, and K.-L. Tsui, "Intelligent railway foreign object detection: A semi-supervised convolutional autoencoder based method," *arXiv preprint arXiv:2108.02421*, 2021.
- [43] Z. Chen, J. Yang, Z. Feng, and H. Zhu, "Railfod23: A dataset for foreign object detection on railroad transmission lines," *Scientific Data*, vol. 11, no. 1, p. 72, 2024.
- [44] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 8759–8768, 2018.
- [45] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2117–2125, 2017.
- [46] R. Khanam and M. Hussain, "Yolov11: An overview of the key architectural enhancements," *arXiv preprint arXiv:2410.17725*, 2024.
- [47] Y. Tian, Q. Ye, and D. Doermann, "Yolov12: Attention-centric real-time object detectors," *arXiv preprint arXiv:2502.12524*, 2025.
- [48] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "Yolox: Exceeding yolo series in 2021," *arXiv preprint arXiv:2107.08430*, 2021.



Zhanyang Wang received his bachelor's degree in Electrical Engineering and Automation from Shandong Jianzhu University in 2022. He is currently pursuing a master's degree in Electrical Engineering at the School of Electrical Engineering, Beijing Jiaotong University. His research interests include computer vision and intelligent detection.



Xin Du received her Ph.D. degree in Control Theory and Control Engineering from the Institute of Automation, Graduate School of the Chinese Academy of Sciences, in 2004. She is currently an Associate Professor at the School of Electrical Engineering, Beijing Jiaotong University. She has presided over and participated in multiple projects funded by the National Natural Science Foundation of China and other research programs. From 2013 to 2014, she was a visiting scholar at the College of Engineering, University of Wisconsin, USA. Her main research interests include power electronics and energy conversion, intelligent detection and control technologies, and their applications in rail transit systems.